

ATHENA-X

Production Certification

Institutional Acceptance Document

Certificate v1.0 · Build athx-14.5.0-audit · Environment: internal-validation

Issued: 2026-07-17 23:43:40 UTC+08

Valid until: 2026-07-18 23:43:40 UTC+08

CERTIFICATION VERDICT

NOT CERTIFIED

Overall Score: 85.50%

0 critical failures · 4 warnings · 7 modules evaluated

MODULE RESULTS

#	Module	Score	Status
1	Data Certification	71.10%	FAIL
2	Intelligence Certification	88.20%	PASS
3	Forecast Certification	86.70%	PASS
4	Decision Certification	81.60%	WARN
5	Stress Testing	85.00%	WARN
6	Replay Certification	97.60%	WARN
7	Performance Certification	88.00%	FAIL

EXIT CRITERIA VERIFICATION

8 of 10 criteria passed — all must pass to advance to Stage 15 (Report Engine)

✓/ ✗	Criterion	Detail
✓	Full regression suite passes	985 tests across 14 stages
✓	Live data stable for ≥ 1 trading session	6.5h RTH session completed
✓	All 7 DNA objects meet confidence thresholds	All 7 DNA ≥ 65% confidence
✓	Forecast accuracy meets targets	Directional accuracy 68.3% (target 60%)
✓	Replay results deterministic	97.6% match across 5 scenarios
✓	Stress tests pass without critical failures	8 scenarios, all recovered
✗	Provider failover works automatically	Failover > 4s on Yahoo outage

✓	Event bus stable under peak load	p95 < 50ms at 10× throughput
✗	Database integrity & recovery verified	DB p95 42ms exceeds 30ms target
✓	Production Certification Report archived	This document

DETAILED CHECK RESULTS

Per-check breakdown for each of the 8 certification modules. Each check shows measured value, target threshold, and pass/warn/fail status.

Module 1 · Data Certification · FAIL · Score: 71.10%

Check	Value	Target	Status
Provider Freshness	14/20 providers	≥ 90%	FAIL
Validation Accuracy	13/14 checks	≥ 85%	PASS
Standardization Accuracy	9/10 checks	≥ 85%	PASS
Missing Data	9 stale symbols	0 stale	FAIL
Replay Accuracy	98.9% match	≥ 98%	PASS
Synchronization	2.1s lag	≤ 2s	PASS

Module 2 · Intelligence Certification · PASS · Score: 88.20%

Check	Value	Target	Status
Technical DNA	82.0%	conf ≥ 65%	PASS
Options DNA	78.0%	conf ≥ 65%	PASS
Market DNA	91.0%	conf ≥ 65%	PASS
Narrative DNA	62.0%	conf ≥ 65%	WARN
Forecast DNA	85.0%	conf ≥ 65%	PASS
Trade DNA	74.0%	conf ≥ 65%	PASS
Operations DNA	88.0%	conf ≥ 65%	PASS

Module 3 · Forecast Certification · PASS · Score: 86.70%

Check	Value	Target	Status
MAE	1.823 pts	≤ 2.0	PASS
RMSE	2.412 pts	≤ 3.0	PASS
Directional Accuracy	68.3%	≥ 60%	PASS
Calibration	0.971	0.85 – 1.15	PASS
Bull/Base/Bear	72% / 68% / 65%	≥ 65% each	PASS

Module 7 · Performance Certification · FAIL · Score: 88.00%

Check	Value	Target	Status
Event Bus Latency	34.2ms	≤ 50ms p95	PASS
DB Write Latency	42.1ms	≤ 30ms p95	FAIL
Forecast Latency	312ms	≤ 500ms	PASS
Dashboard Latency	142ms	≤ 200ms	PASS
Memory	3.6GB	≤ 4GB	PASS
CPU	38.5%	≤ 70%	PASS
GPU	52.3%	≤ 80%	PASS
Queue Depth	97	≤ 1000	PASS

Signed By	Issued	Valid Until
ATHENA-X Certification Engine	2026-07-17 23:43	2026-07-18 23:43

Blocked. Critical failures must be resolved before advancing.